

Garbage Classification Final Report

1. Overview

Topic 8: Image-based waste classification for a greener environment.

The project classifies images into 6 classes: cardboard, glass, metal, paper, plastic, trash.

2. Files added

- preprocess_garbage_images.py
- train_cnn_baseline.py
- train_transfer_learning.py
- evaluate_garbage_classifier.py
- gradcam_visualization.py
- error_analysis.md
- metrics_summary.csv
- wandb_report_link.txt
- demo_app.py
- demo_notebook.ipynb

3. Notes

- The core project scripts are in src/*.py and app.py.
- The report is based on the best checkpoint from outputs/checkpoints/best.pt.
- Streamlit demo can be run with: `python -m streamlit run app.py`